

LABORATORY TECHNIQUES

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CHROMATOGRAPHY

Definition:

Chromatography is a technique used to separate, identify, and quantify components of a mixture based on how they distribute between a mobile phase (liquid or gas) and a stationary phase (solid or liquid).

Key Principle:

- Different substances move at different rates through the stationary phase, causing separation.
- Main Types:
- Gas Chromatography (GC) – mobile phase: gas, for volatile compounds
- Liquid Chromatography (LC) – mobile phase: liquid, for many chemicals and biomolecules
- Thin-Layer Chromatography (TLC) – stationary phase: coated plate, simple and fast separation

APPLICATIONS

- Toxicology: Detect drugs and steroids
- Environmental Testing: Detect pollutants
- Biochemistry: Separate proteins, enzymes, DNA/RNA
- Forensics: Identify drugs, poisons, toxins

BIOCHEMISTRY ANALYZER

- Definition:
- A biochemistry analyzer is a laboratory instrument that measures chemical substances in body fluids (blood, serum, plasma, urine, CSF) to assess metabolic and physiological functions.

PRINCIPLE:

- Operates on spectrophotometry: measures light absorbed by a sample when it reacts with specific reagents.
- Beer-Lambert law: Absorption of light is proportional to the concentration of the substance and thickness of the sample.

FUNCTIONS

- Determines concentrations of glucose, enzymes, hormones, electrolytes, and other analytes accurately and efficiently.